Contribution Journal – Module 10 Lab: Unsupervised LearningIn this module's lab, I implemented two core unsupervised learning techniques—K-Means Clustering and Principal Component Analysis (PCA)—using Python and scikit-learn in Google Colab. First, I worked with synthetic customer data (make_blobs) and applied the Elbow Method by iterating through cluster values from $k=1$ to 10 and plotting the resulting inertia to identify $k=4$ as the optimal number of clusters. After fitting the K-Means model with four clusters, I extracted the cluster labels and centroid coordinates (cluster_centers_) to visualize the segmented groups with matplotlib. Next, I applied PCA to the 4-dimensional Iris dataset, reducing it to 2 principal components while retaining 97.77% of the total variance (92.46% in Component 1 and 5.31% in Component 2), and plotted the compressed data using seaborn. Finally, I debugged formatting and syntax issues in the starter notebook, imported missing libraries, completed the knowledge check questions analyzing the trade-offs of inertia and feature redundancy, and uploaded the completed notebook to my GitHub repository.AI Disclosure (if needed for your submission):What AI was used: Google Gemini.Why it was used: To help troubleshoot syntax errors caused by single-line formatting in the starter notebook cells and to structure the code blocks for K-Means and PCA.What I learned in reflection: I learned how the Elbow Method balances low inertia against overfitting when selecting k, and how capturing 97.77% of variance in just two PCA components proves that the original four Iris features are highly correlated.How I modified it to make it my own: I executed and verified each code block in my Google Colab environment, added the missing seaborn import to generate the PCA visualization, analyzed the exact variance output from my runtime, and committed the final .ipynb file to my GitHub repository.